CUSTOMER: 447526	ORDER #: 2062971
BATCH #: 1751371	DELIVERY ID: 13091299
PICK ZONE: DRUM	PICK SEQUENCE #: 2400
PRODUCT NAME: MB-215/ANCOCIDE 4070	BARCODE #: 10189589

Material Safety Data Sheet: MB-215/ANCOCIDE 4070

Supercedes Date 12/13/2011 **Issuing Date** 07/09/2013

1. PRODUCT AND COMPANY IDENTIFICATION

Product Name MB-215/ANCOCIDE 4070
Recommended use Biocidal product
Information on Manufacturer
CHEM-AQUA, INC
BOX 152170
IRVING, TEXAS 75015

Product Code 10189589
Chemical nature Aqueous solution
Emergency Telephone Number
CHEMTREC® 800-424-9300

2. HAZARDS IDENTIFICATION

Emergency Overview
DANGER
Corrosive
Causes skin and eye burns
May cause allergic skin reaction
May cause delayed lung injury and burns
Harmful or fatal if swallowed

Color Colorless - Yellow **Physical State** Liquid **Odor** Pungent

Potential Health Effects	
Principle Route of Exposure	Skin contact, Eye contact, Inhalation.
Primary Routes of Entry	Skin Absorption
Acute Effects	
Eyes	Corrosive to the eyes and may cause severe damage including blindness.
Skin	Causes skin burns. May cause allergic skin reaction.
Inhalation	Harmful by inhalation. Causes burns. Risk of serious damage to the lungs (by inhalation).
Ingestion	If ingested, severe burns of the mouth and throat, as well as a danger of perforation of the esophagus and the stomach. Components of the product create formation of methemoglobin.
Chronic Toxicity	May cause sensitization by skin contact. Inhaled corrosive substances can lead to a toxic edema of the lungs.
Target Organ Effects	Immune system, Blood.
Aggravated Medical Conditions	Skin disorders, Respiratory system.
Potential Environmental Effects	See Section 12 for additional Ecological information.

3. COMPOSITION / INFORMATION ON INGREDIENTS

Component	CAS-No
Magnesium nitrate	10377-60-3
5-Chloro-2-methyl-4-isothiazolin-3-one	26172-55-4
Magnesium chloride	7786-30-3

4. FIRST AID MEASURES

General advice	Do not get in eyes, on skin or on clothing. Do not breathe mist.
Eye Contact	Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Remove contact lenses. Call a physician or poison control center immediately.
Skin Contact	Remove immediately all contaminated clothing. Wash off immediately with plenty of water for at least 15 minutes. Contact a poison control center.
Inhalation	Move to fresh air. If not breathing, give artificial respiration. Get medical attention immediately.
Ingestion	Call a physician or poison control center immediately. Give small amounts of water to drink. DO NOT induce vomiting unless directed to do so by a physician or poison control center.
Notes to physician	The product causes burns of eyes, skin and mucous membranes. Control of circulatory system, shock therapy if needed. Since reversion of methemoglobin to hemoglobin occurs spontaneously after termination of exposure, moderate degrees of cyanosis need to be treated only by supportive measures. May cause sensitization of susceptible persons.

5. FIRE-FIGHTING MEASURES

Flash Point Does not flash **Method** Not applicable
Autoignition Temperature No information available.
Flammability Limits in Air % Hydrogen, by reaction with metals. **Upper** 75 **Lower** 4
Suitable Extinguishing Media
Foam. Alcohol-resistant foam. Carbon dioxide (CO2). Dry chemical. Water spray. Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.
Specific hazards arising from the chemical
Thermal decomposition can lead to release of irritating gases and vapors. Contact with metals liberates flammable hydrogen gas. Material can create slippery conditions.
Protective Equipment and Precautions for Firefighters
As in any fire, wear self-contained breathing apparatus pressure -demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

NFPA	**Health** 3	**Flammability** 1	**Instability** 0
HMIS	**Health** 3	**Flammability** 1	**Instability** 0

6. ACCIDENTAL RELEASE MEASURES

Personal Precautions Use personal protective equipment. Ensure adequate ventilation. Prevent further leakage or spillage if safe to do so. Material can create slippery conditions.
Environmental Precautions Do not flush into surface water or sanitary sewer system.
Methods for Containment Contain spillage, soak up with non -combustible absorbent material, (e.g. sand, earth, diatomaceous earth, vermiculite) and transfer to a container for disposal according to local / national regulations (see section 13)
Methods for Cleaning Up Pick up and transfer to properly labeled containers.
Neutralizing Agent Neutralize with lime milk or soda and flush with plenty of water.

7. HANDLING AND STORAGE

Handling Ensure adequate ventilation. Wear personal protective equipment.
Storage Keep out of the reach of children. Store in original container. Keep containers tightly closed in a cool, well-ventilated place. Metal containers must be lined. Freezing will affect the physical condition but will not damage the material. Thaw and mix before using.
Storage Temperature **Minimum** 34 °F / 1 °C **Maximum** 131 °F / 55 °C
Storage Conditions **Indoor** X **Outdoor** **Heated** **Refrigerated**

8. EXPOSURE CONTROLS / PERSONAL PROTECTION

Exposure Guidelines

Component	ACGIH TLV	OSHA PEL	NIOSH
Magnesium nitrate	No data available	No data available	No data available
5-Chloro-2-methyl-4-isothiazolin-3-one	No data available	No data available	No data available
Magnesium chloride	No data available	No data available	No data available

Engineering Measures Ensure adequate ventilation, especially in confined areas. Where reasonably practicable this should be achieved by the use of local exhaust ventilation and good general extraction.
Personal Protective Equipment
Eye/Face Protection Tightly fitting safety goggles. Face-shield.
Skin Protection Wear suitable protective clothing, Impervious gloves.
Respiratory Protection In case of inadequate ventilation wear respiratory protection. When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.
General Hygiene Considerations Wear protective gloves/clothing. Ensure that eyewash stations and safety showers are close to the workstation location. Remove and wash contaminated clothing before re -use.

9. PHYSICAL AND CHEMICAL PROPERTIES

Physical State	Liquid	**Viscosity**	Non viscous
Color	Colorless - Yellow	**Odor**	Pungent
Appearance	Transparent	**pH**	2
Specific Gravity	1.02	**Evaporation Rate**	0.58 (Butyl acetate=1)
Percent Volatile (Volume)	97.9	**VOC Content (%)**	0
VOC Photoreactive (Y/N)	No	**VOC Content (g/L)**	0
Vapor Pressure	17.05 mmHg @ 70°F	**Vapor Density**	0.6 (Air = 1.0)
Solubility	Completely soluble	**Boiling Point/Range**	212 °F / 100 °C

10. STABILITY AND REACTIVITY

Chemical Stability Stable. Hazardous polymerization does not occur.

Conditions to Avoid	None known
Incompatible Products	Strong oxidizing agents, Reducing agents, Amines, Powdered metals, Light and/or alkaline metals, Contact with metals liberates hydrogen gas.
Hazardous Decomposition Products	Carbon oxides, Nitrogen oxides (NOx), Sulfur oxides, Hydrogen chloride gas.
Possibility of Hazardous Reactions	None under normal processing

11. TOXICOLOGICAL INFORMATION

Product Information No information available.

Component Information

Acute Toxicity

Component	LD50 Oral	LD50 Dermal	LC50 Inhalation	Draize Test	Other
Magnesium nitrate	= 5440 mg/kg (Rat)	no data available	no data available	no data available	no data available
5-Chloro-2-methyl-4-isothiazolin-3-one	= 481 mg/kg (Rat)	> 1008 mg/kg (Rat)	= 1.23 mg/L (Rat) 4 h	no data available	no data available
Magnesium chloride	= 2800 mg/kg (Rat)	no data available	no data available	no data available	no data available

Chronic Toxicity

Component	Mutagenicity	Sensitization	Developmental Toxicity	Reproductive Toxicity	Target Organ Effects
Magnesium nitrate	no data available	no data available	no data available	no data available	no data available
5-Chloro-2-methyl-4-isothiazolin-3-one	no data available	Skin sensitization	no data available	no data available	Immune system
Magnesium chloride	no data available	no data available	no data available	no data available	no data available

Carcinogenicity

Component	ACGIH	IARC	NTP	OSHA	Other
Magnesium nitrate	not applicable	Group 2A	not applicable	X	not applicable
5-Chloro-2-methyl-4-isothiazolin-3-one	not applicable	not applicable	not applicable	not applicable	not applicable
Magnesium chloride	not applicable	not applicable	not applicable	not applicable	not applicable

12. ECOLOGICAL INFORMATION

Product Information No information available.

Component Information

Component	Toxicity to Algae	Toxicity to Fish	Microtox	Water Flea	log Pow
Magnesium nitrate	no data available	no data available	no data available	no data available	N/A
5-Chloro-2-methyl-4-isothiazolin-3-one	EC50 = 0.31 mg/L Anabaena flos-aquae 120 h EC50 0.11 - 0.16 mg/L Pseudokirchneriella subcapitata 72 h EC50 0.03 - 0.13 mg/L Pseudokirchneriella subcapitata 96 h	LC50 = 1.6 mg/L Oncorhynchus mykiss 96 h	EC50 = 5.7 mg/L 16 h	EC50= 4.71 mg/L 48 h EC50 0.12 - 0.5 mg/L 48 h EC50 0.71 - 0.99 mg/L 48 h	-0.71 - 0.75
Magnesium chloride	EC50 = 2200 mg/L Desmodesmus subspicatus 72 h	LC50 = 4210 mg/L Gambusia affinis 96 h LC50 1970 - 3880 mg/L Pimephales promelas 96 h	EC50 = 26140 mg/L 1 h EC50 = 36300 mg/L 30 min EC50 = 77200 mg/L 24 h	EC50= 1400 mg/L 24 h EC50= 140 mg/L 48 h	N/A

Persistence and Degradability	No information available.
Bioaccumulation	No information available.
Mobility	No information available.

13. DISPOSAL CONSIDERATIONS

Product Disposal	Pesticide wastes are acutely hazardous. Improper disposal of excess pesticide or rinsate is a violation of federal law. If these wastes cannot be disposed of by use according to label instructions, contact your state pesticide or environmental control agency.
Container Disposal	Empty containers should be taken for local recycling, recovery, or waste disposal. Do not re -use empty containers.

14. TRANSPORT INFORMATION

DOT

Proper Shipping Name	Corrosive liquid, acidic, organic, n.o.s.
Hazard Class	8
UN-No	UN3265
Packing Group	II
Description	Corrosive liquid, acidic, organic, n.o.s.(5 -chloro-2-methyl-4-isothiazolin-3-one),8,UN3265,PG II

TDG

Proper shipping name	Corrosive liquid, acidic, organic, n.o.s.
Hazard Class	8
UN-No	UN3265
Packing Group	II
Description	Corrosive Liquid, Acidic, Organic, N.O.S.,.(5 -chloro-2-methyl-4-isothiazolin-3-one),8,UN3265,PG II

ICAO

UN-No	UN3265
Proper Shipping Name	Corrosive liquid, acidic, organic, n.o.s.*
Hazard Class	8
Packing Group	II
Shipping Description	Corrosive liquid, acidic, organic, n.o.s.*(5 -chloro-2-methyl-4-isothiazolin-3-one),8,UN3265,PG II

IATA

UN-No	UN3265
Proper Shipping Name	Corrosive liquid, acidic, organic, n.o.s.*
Hazard Class	8
Packing Group	II
ERG Code	8L
Shipping Description	UN3265,Corrosive liquid, acidic, organic, n.o.s.*(5 -chloro-2-methyl-4-isothiazolin-3-one),8,PG II

IMDG/IMO

Proper Shipping Name	Corrosive liquid, acidic, organic, n.o.s.
Hazard Class	8
UN-No	UN3265
Packing Group	II
EmS No.	F-A, S-B
Shipping Description	UN3265, Corrosive liquid, acidic, organic, n.o.s.(5 -chloro-2-methyl-4-isothiazolin-3-one,8,PG II

15. REGULATORY INFORMATION

Inventories

TSCA	Complies
DSL	Complies

U.S. Federal Regulations

SARA 313

Section 313 of Title III of the Superfund Amendments and Reauthorization Act of 1986 (SARA). This product contains a chemical or chemicals which are subject to the reporting requirements of the Act and Title 40 of the Code of Federal Regulations, Part 372

Component	CAS-No	Weight %	SARA 313 - Threshold Values
Magnesium nitrate	10377-60-3	1-5	1.0

SARA 311/312 Hazardous Categorization

Acute Health Hazard	Chronic Health Hazard	Fire Hazard	Sudden Release of Pressure Hazard	Reactive Hazard
Yes	Yes	No	No	No

CERCLA

Component	Hazardous Substances RQs	CERCLA EHS RQs
Magnesium nitrate	Not applicable	Not applicable
5-Chloro-2-methyl-4-isothiazolin-3-one	Not applicable	Not applicable
Magnesium chloride	Not applicable	Not applicable

Canada

This product may not be commercially placed on the market in Canada.

WHMIS Hazard Class

Not applicable

16. OTHER INFORMATION

Prepared By Rachael Mohochi

Supercedes Date	12/13/2011
Issuing Date	07/09/2013
Reason for Revision	No information available.
Glossary	No information available.
List of References.	No information available.

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